Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2025

Course: B.Tech
Code: 104501

Subject: Computer Networks and Security

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In computer network which of the following is an example of guided transmission media?
(i) Radio waves
(ii) Microwave
(iii) Optical fiber
(iv) Infrared

(b) Which techniques is used in Cyclic Redundancy Check?
(i) XOR Encryption
(ii) AND operation
(iii) OR operation
(iv) Modulo-2 Division

(c) Multiple access protocol which is used carrier sensing and collision detection?
(i) CSMA/CD
(ii) CDMA/CA
(iii) Pure ALOHA
(iv) Slotted ALOHA

(d) What is the primary function of packet forwarding at the network layer?
(i) Delivering packets to their next hop on the path to the destination
(ii) Assigning IP addresses
(iii) Encrypting data
(iv) Fragmenting packets

(e) One of the features that is provided by TCP but not by UDP?
(i) Error detection
(ii) Connection-oriented communication
(iii) Broadcasting
(iv) Fast delivery

(f) Which protocol is used for monitoring and managing network devices?
(i) DNS
(ii) SNMP
(iii) SMTP
(iv) FTP

(g) A hash function takes an input message and returns.....
(i) a fixed-size string of characters
(ii) a variable-size string of characters
(iii) input size string characters
(iv) a public-private key pair

(h) What is the main idea behind the RSA public-key encryption algorithm?
(i) Using a single symmetric key
(ii) Hiding the IP address of the sender
(iii) Using large prime numbers and modular arithmetic
(iv) Using XOR operations on data

(i) The primary function of a Virtual Private Network is?
(i) Creates a secure tunnel over a public network for private communication
(ii) Acts as a physical router
(iii) Assigns MAC addresses dynamically
(iv) Encrypts DNS records only

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(j) In SSL architecture, which protocol is responsible for managing encryption, compression, and message integrity?

(i) SSL Handshake Protocol
(ii) SSL Alert Protocol
(iii) SSL Record Protocol
(iv) Change Cipher Spec Protocol

Q.2 (a) Describe various network topologies used in computer networks. [7]
(b) Discuss Frequency Division Multiplexing, Time Division Multiplexing, Wavelength Division Multiplexing. [7]

Q.3 (a) Explain its features and advantages over TCP and UDP. Discuss the concept of congestion control in SCTP. [7]
(b) Illustrate Logical addressing – IPv4, IPv6. [7]

Q.4 (a) Differentiate between wired LAN and wireless LAN. Discuss the concept a Virtual LAN. [7]
(b) Explain the OSI reference model in detail. How it is different from TCP/IP model. [7]

Q.5 (a) Explain the working principles of Pure ALOHA, Slotted ALOHA, CSMA/CD, and CSMA/CA with their applications. [7]
(b) Explain the Sliding Window Protocol and Piggybacking technique. How do these methods improve data transmission efficiency in computer networks? [7]

Q.6 (a) What is the size of an ARP packet when the protocol is IP and the hardware is Ethernet? [7]
(b) Discuss Three-ways Handshaking in a TCP connection. [7]

Q.7 (a) Define Internet Protocol. Is the size of the ARP packet fixed? Explain. [7]
(b) An IP datagram must go through router 128.46.10.5. There are no other restrictions on the routers to be visited. Draw the IP options with their values. [7]

Q.8 (a) Discuss Digital Signatures. Explain Secret-key signature and Public-key signature. [7]
(b) The Diffie-Hellman key exchange is being used to establish a secret key between Alice and Bob. Alice send Bob (719, 3, 191). Bob respond with (543). Alice's secret key, x, is 16. Explain the secret key with the help of above example. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]
(a) Token Bucket algorithm
(b) File Transfer Protocol
(c) Secure Socket Layer and Transport layer Security
(d) Error control protocols

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